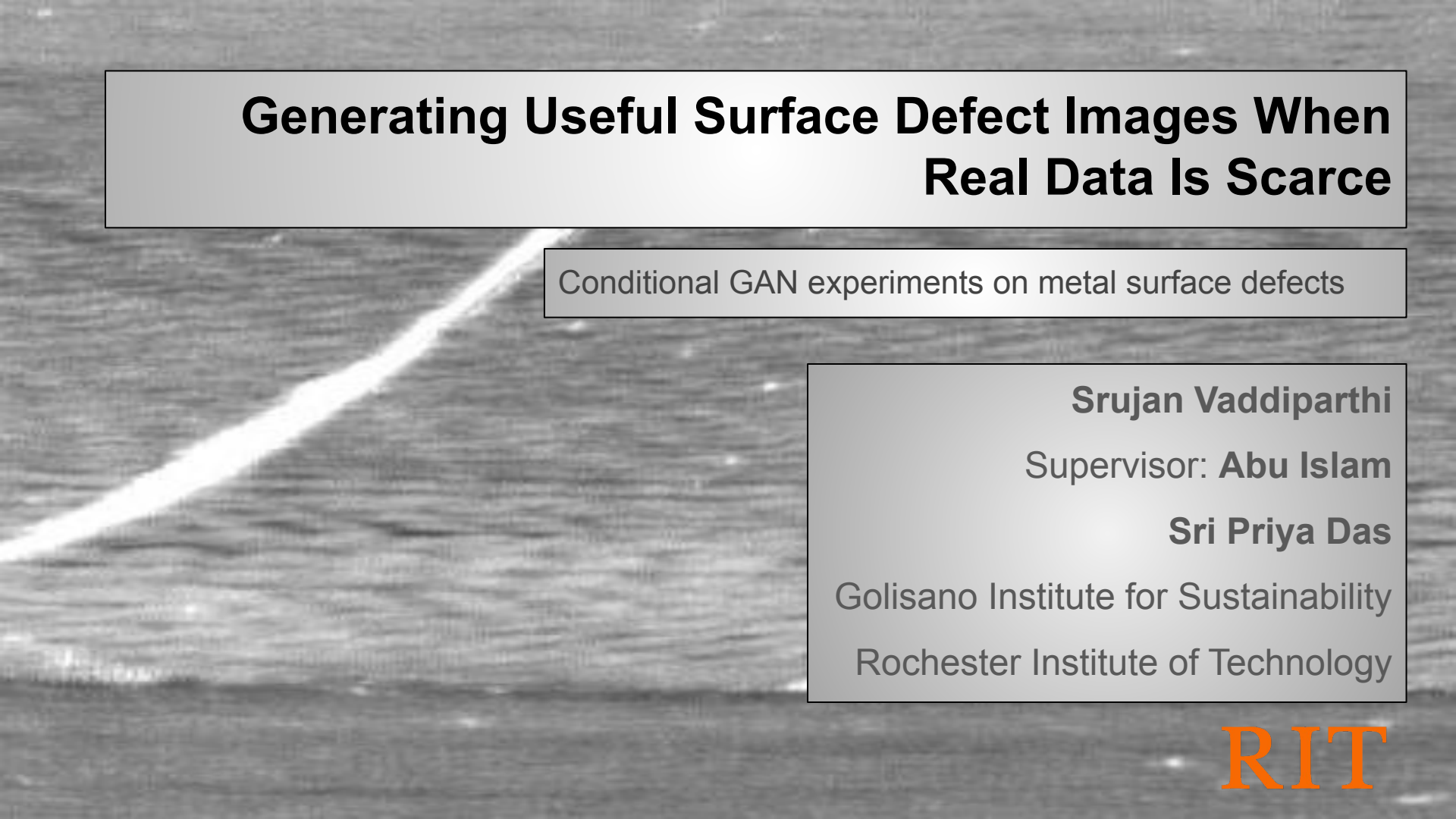




Generating Useful Surface Defect Images When Real Data Is Scarce

Conditional GAN experiments on metal surface defects

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Golisano Institute for Sustainability

Rochester Institute of Technology

RIT

Agenda

- Why this problem matters
- Research question
- How a conditional GAN works
- Experimental progression and results
- Key lessons, challenges, and next steps

Why this problem matters

- Defect images are often scarce in industrial settings

Why this problem matters

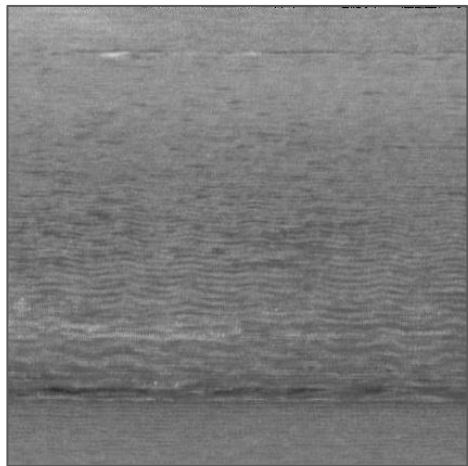
- Defect images are often scarce in industrial settings
- Small datasets limit computer vision performance

Why this problem matters

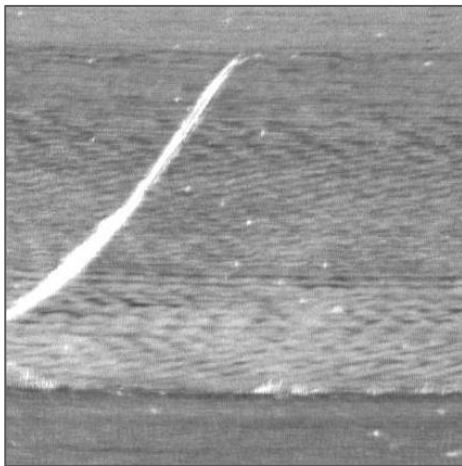
- Defect images are often scarce in industrial settings
- Small datasets limit computer vision performance
- Useful synthetic images could support inspection, reuse, and reduced waste

Why this problem matters

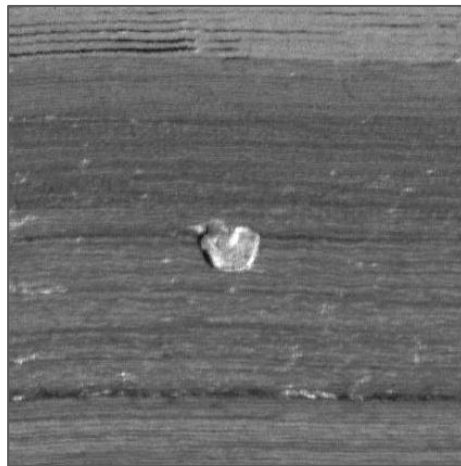
- Defect images are often scarce in industrial settings
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Normal



Scratch



Spot

This is not a general image-generation problem. It is a low-data, defect-faithful industrial generation problem.

Research Question

Can AI generate useful defect images when real surface-defect data is scarce?

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Research Question

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What I wanted to understand:

- Can the model generate the defect type I ask for?
- Which defect types are easier to learn?

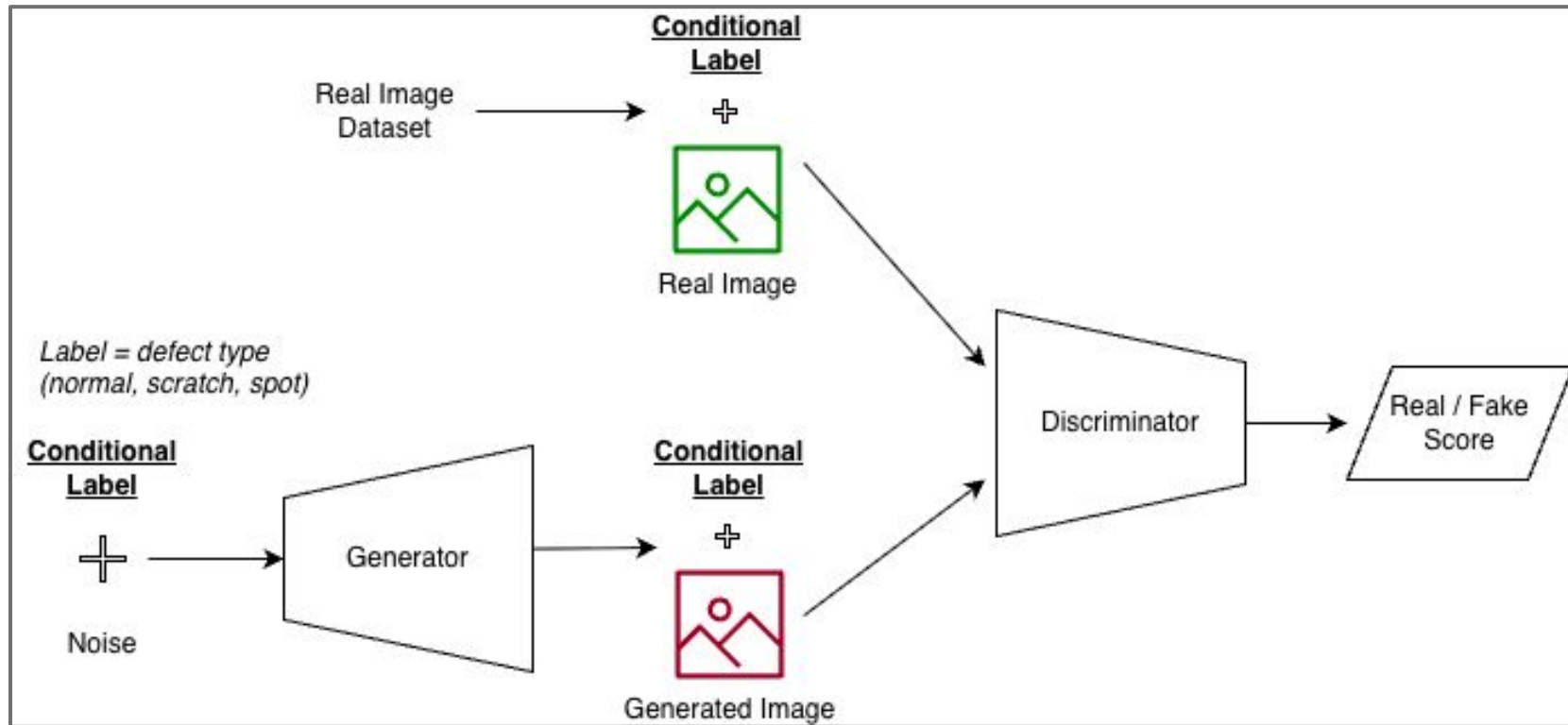
Research Question

Can AI generate useful defect images when real surface-defect data is scarce?

What I wanted to understand:

- Can the model generate the defect type I ask for?
- Which defect types are easier to learn?
- What improves output quality when data is both scarce and imbalanced?

How a Conditional GAN Works (Simplified)



The label steers what the generator creates and what the discriminator expects.

How I Judged Progress

How I evaluated outputs:

- Visual Realism 
- Class Consistency 
- Diversity 

Starting Point: 3-Class Baseline

- One cDCGAN for normal / scratch / spot

Starting Point: 3-Class Baseline

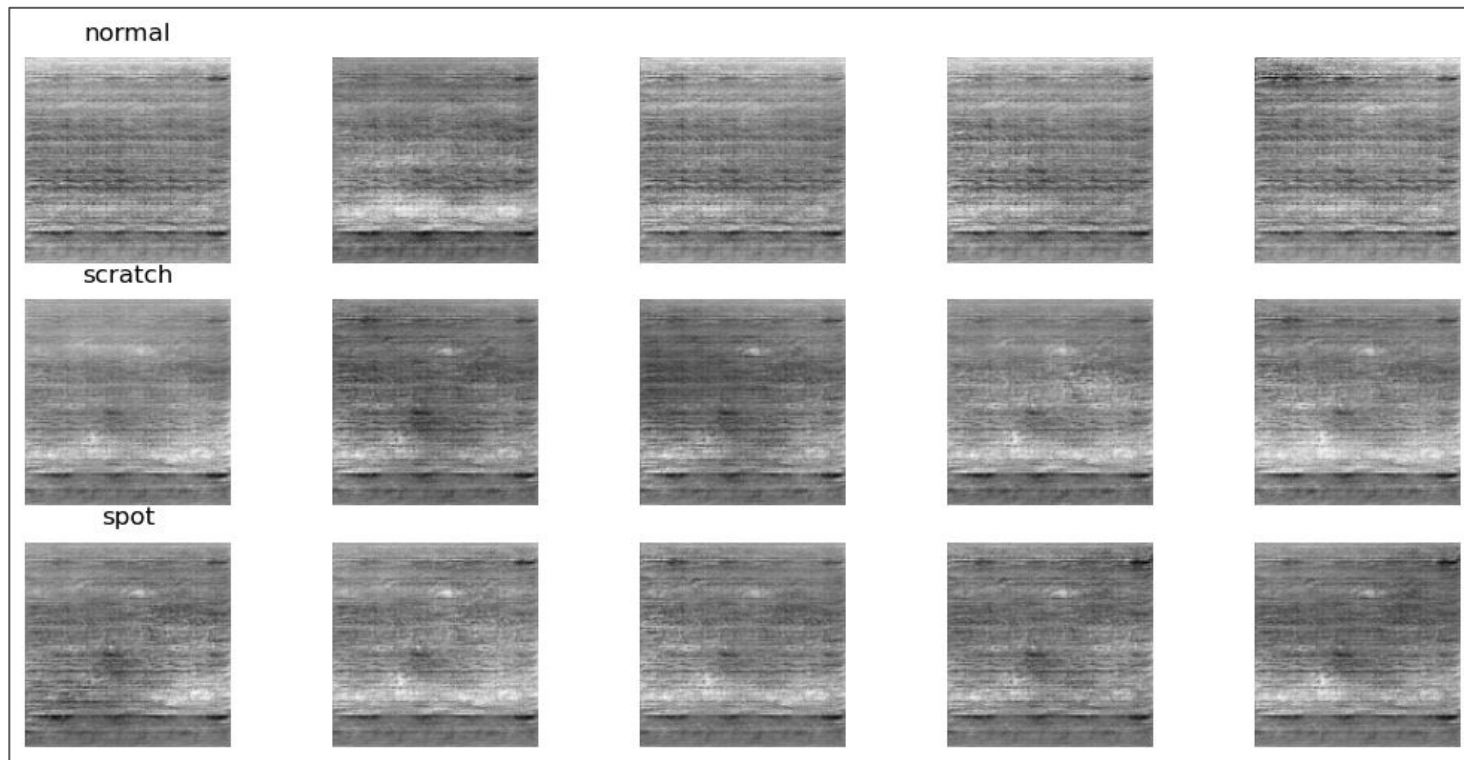
- One cDCGAN for normal / scratch / spot
- Learned shared metal texture

Starting Point: 3-Class Baseline

- One cDCGAN for normal / scratch / spot
- Learned shared metal texture
- Weak defect-specific generation

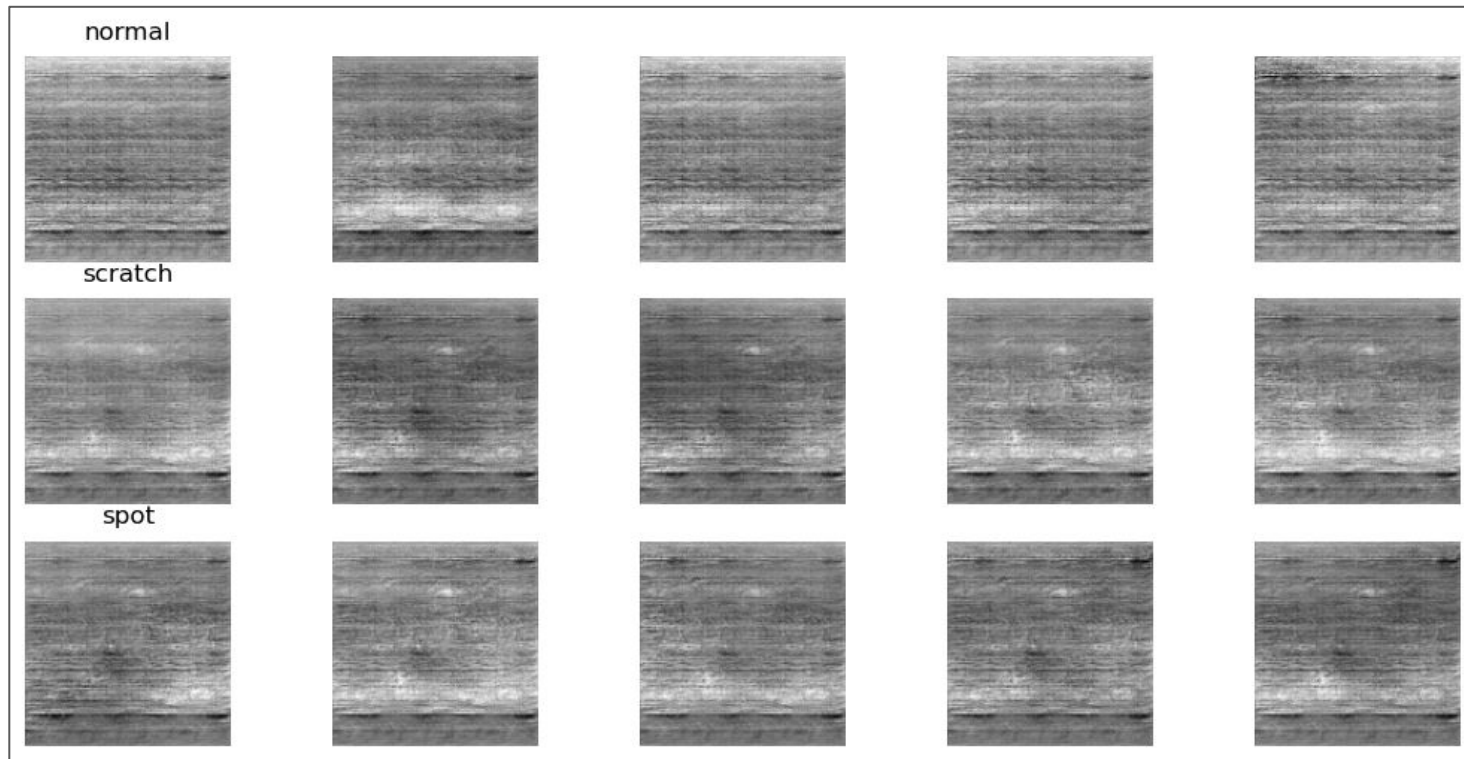
Starting Point: 3-Class Baseline

Generated Output:



Starting Point: 3-Class Baseline

Generated Output:



*The 3-class setup
learned metal
texture better than
defect structure.*

Why I Simplified the Problem

From 3 classes to binary experiments.

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 - **normal vs scratch**
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Why I Simplified the Problem

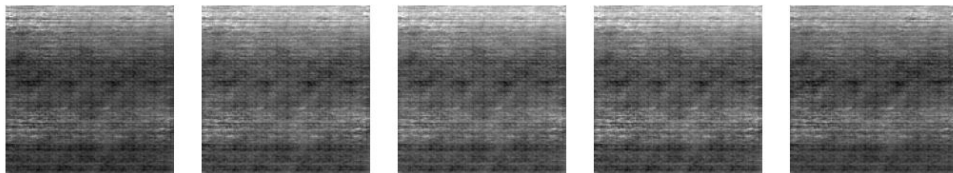
From 3 classes to binary experiments.

- The 3-class setup weakened class-specific generation.
- I split the problem into:
 - **normal vs scratch**
 - **normal vs spot**
- I trained each setup for the same number of epochs, and also checked intermediate results.

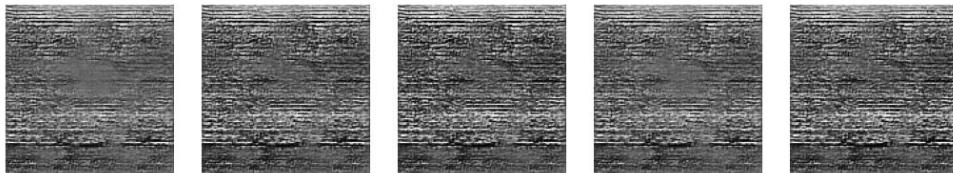
Normal vs Scratch: What Changed

Baseline Output:

normal



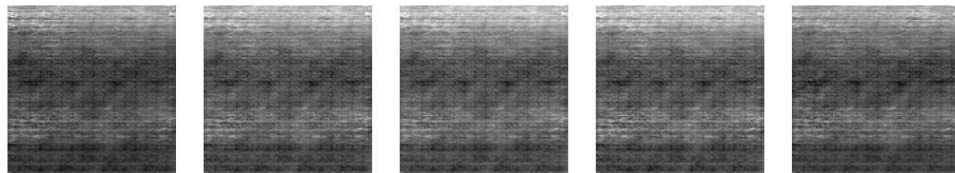
scratch



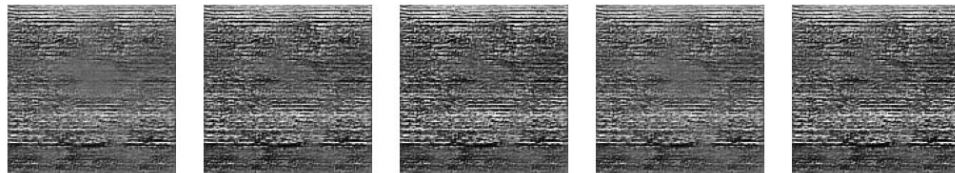
Normal vs Scratch: What Changed

Baseline Output:

normal

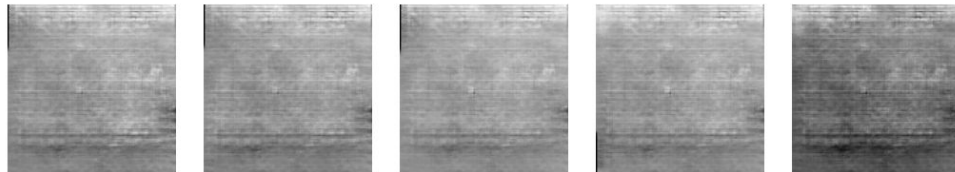


scratch

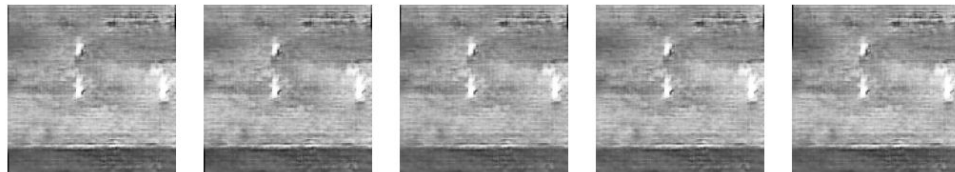


Final Setup Output:

normal



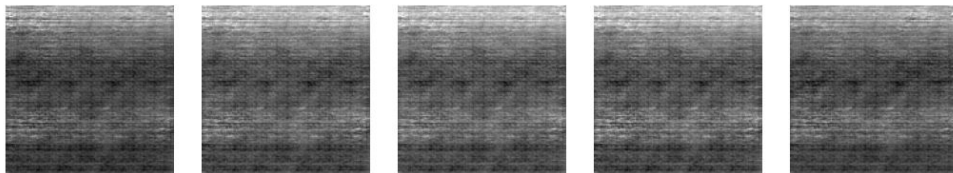
scratch



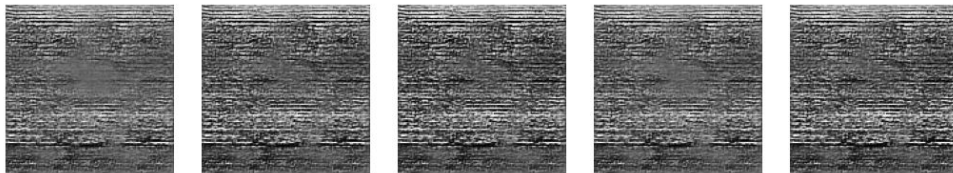
Normal vs Scratch: What Changed

Baseline Output:

normal

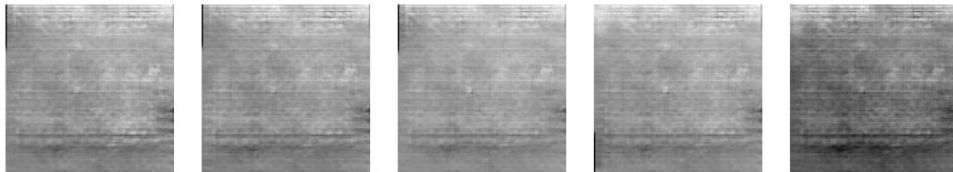


scratch

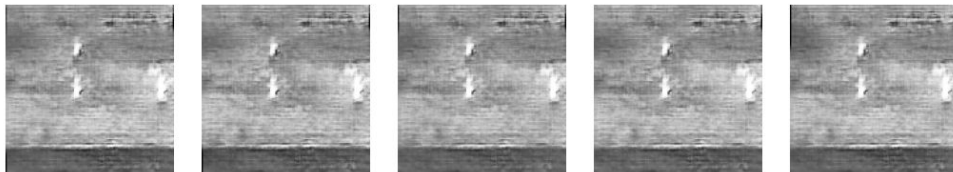


Final Setup Output

normal



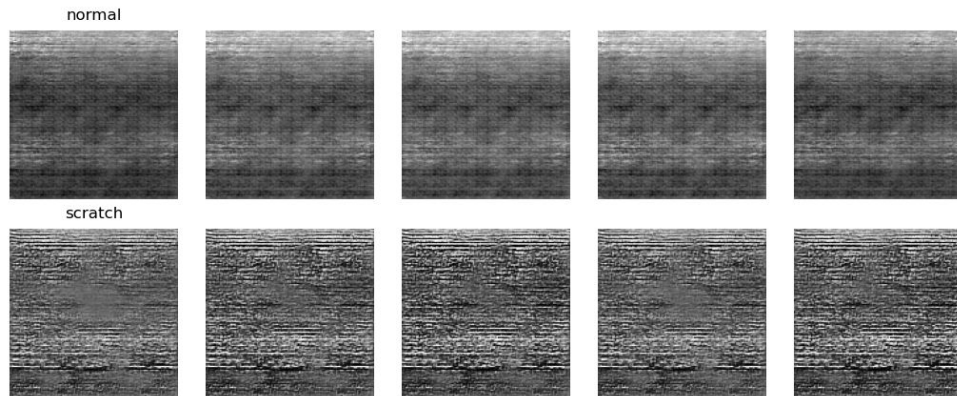
scratch



Evaluation	Judgement
Visual Realism 	Medium
Class Consistency 	Medium
Diversity 	Low

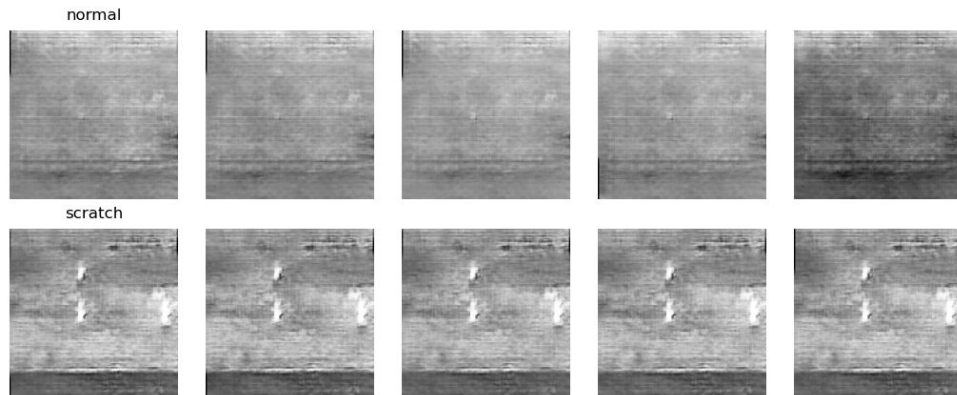
Normal vs Scratch: What Changed

Baseline Output:



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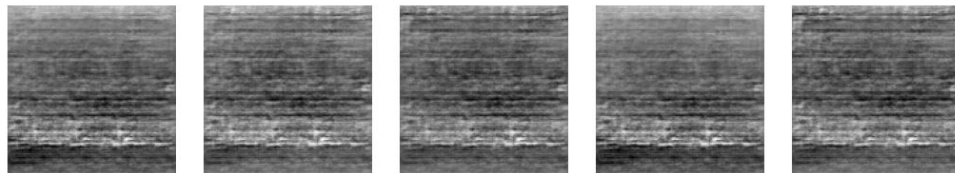


- Binary setup + enhancement improved scratch-vs-normal separation
- Diversity stayed low, and scratches were still hard to model

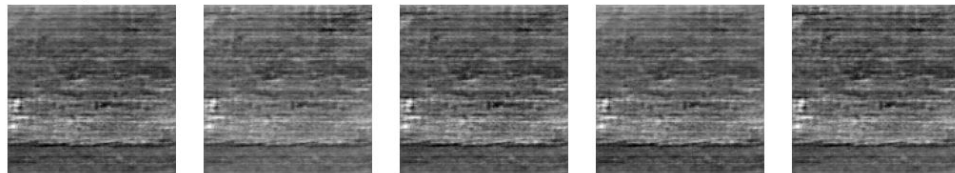
Normal vs Spot: What Changed

Baseline Output:

normal



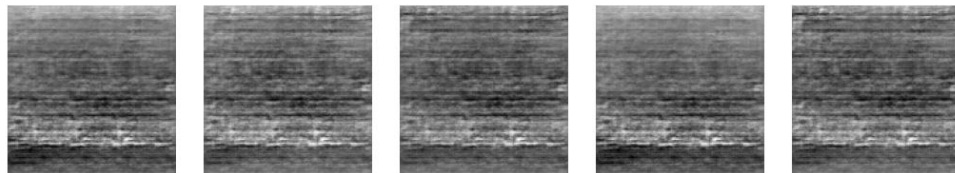
spot



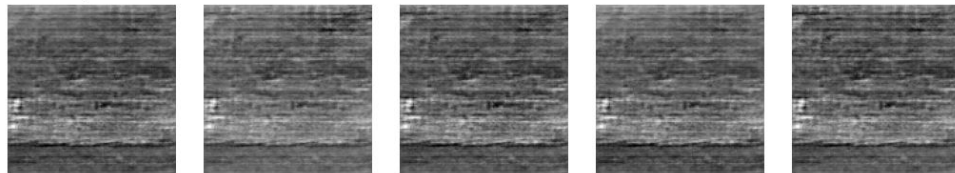
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Baseline Output:

normal

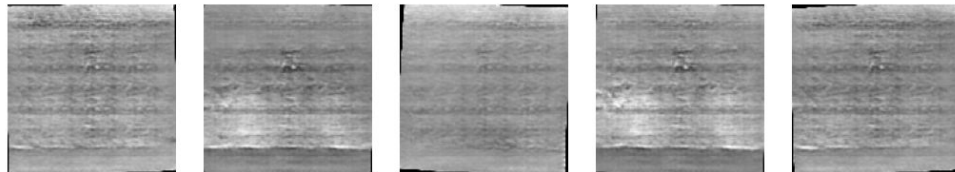


spot

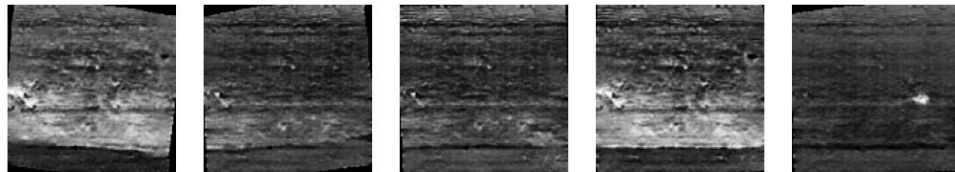


Final Setup Output

normal



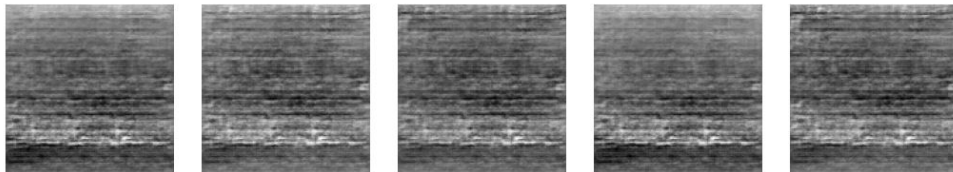
spot



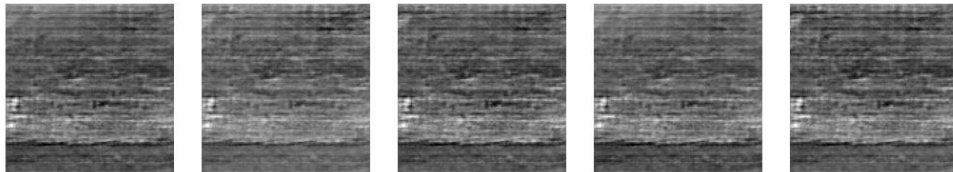
Normal vs Spot: What Changed

Baseline Output:

normal

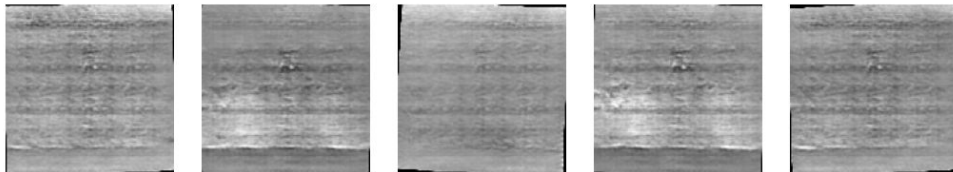


spot

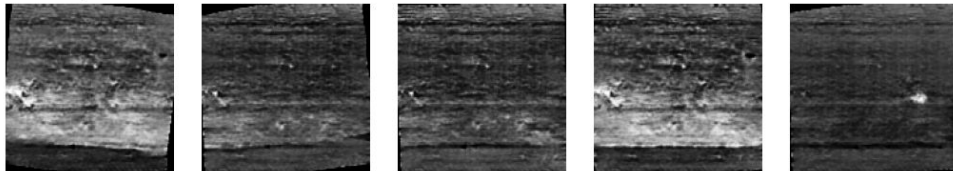


Final Setup Output

normal



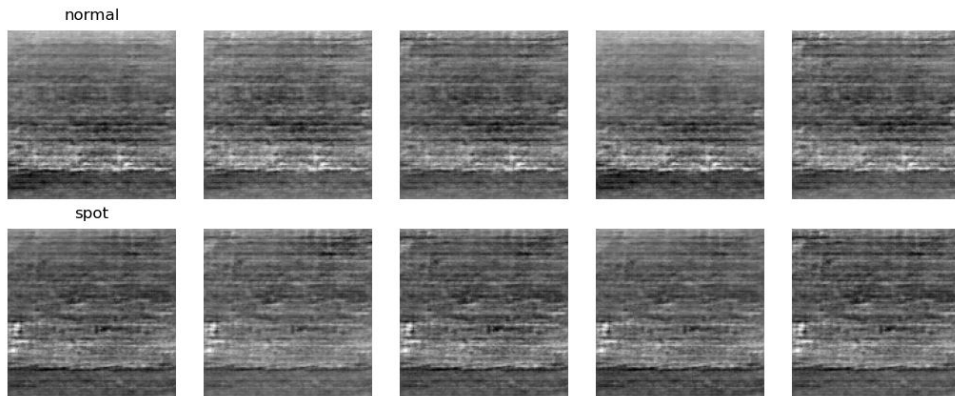
spot



Evaluation	Judgement
Visual Realism 	Medium
Class Consistency 	High
Diversity 	Low

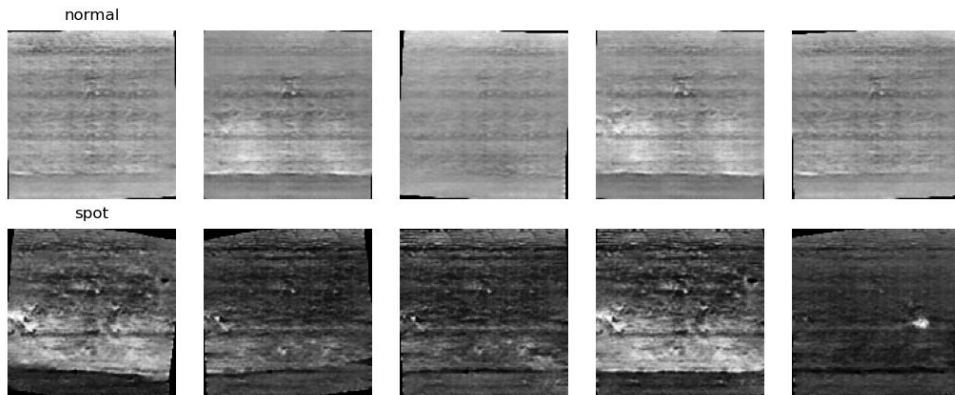
Normal vs Spot: What Changed

Baseline Output:



Evaluation	Judgement
Visual Realism 🦋	Medium
Class Consistency 🏷️	High
Diversity 📊	Low

Final Setup Output



- Spots were easier to learn than scratches
- Class consistency improved, but diversity stayed low

What I learned

- Spot-like defects were easier to generate than scratches
- Repeated examples and limited surface variety strongly shaped what the model learned
- The model could separate classes better, but still repeated similar outputs

Challenges

- **Structured defects were harder than localized ones**
- **Narrow and repeated data reduced diversity**
- **The current cDCGAN learned texture more easily than controlled defect structure**
- **Best-looking outputs sometimes appeared before the final epoch**

What comes next

- **More diverse defect data**
- **Stronger conditioning / better architectures**
- **Image-to-image translation approaches**
- **More reliable ways to identify the best outputs**

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These findings suggest a path from low-data generation experiments toward more controllable and useful defect synthesis.

Conclusion

- The cDCGAN learned metal texture and some class-specific cues
- Spots were easier to learn than scratches
- Diversity remained the main limitation
- These results point clearly to the next set of experiments

References / Acknowledgments

- **Mirza & Osindero — Conditional Generative Adversarial Nets**
- **Radford et al. — DCGAN**
- **Isola et al. — pix2pix**
- **Zhu et al. — CycleGAN**
- **Defect-Transfer GAN paper**
- **Advisor: Abu Islam**
- **Co-mentor / support: Sri Priya Das**
- **RIT Research Computing**

An aerial, black and white photograph of a body of water, likely the ocean. A bright, white, curved line representing a boat's wake stretches from the bottom left towards the upper center of the frame. The water surface is textured with small waves and ripples.

Thank You!

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